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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARENCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCCHE100 -

FUNDAMENTALS OF CHEMISTRY - II

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions, each question carries **1** mark

1. (b) A Lewis acid accept a pair of electrons from a Lewis base
2. (c) SO_3
3. (b) Long range order
4. (c) base
5. (b) $1^o > 2^o > 3^o$
6. (a) N/m
7. (c) 90ml
8. (a) Has more hydrogen atoms
9. (a) PV=nRT
10. (d) 7

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11. Explain the concept of 'oxidation' and 'reduction' in terms of electron transfer.

The oxidation process involves the loss of electrons, whereas the reduction process involves the gain of electrons. Oxidation and reduction are complementary processes. A reaction in which both oxidation and reduction occur simultaneously is known as a redox reaction.

$$Mg \longrightarrow Mg^{2+} + 2e^- \quad \textbf{(oxidation)}$$

$$O + 2e^- \longrightarrow O^{2-} \quad \textbf{(reduction)}$$

(Explanation with relevant examples give full marks)

12. Give an account of nucleophiles.

Nucleophiles (meaning "nucleus-loving") are electron-rich species that are attracted to electron-deficient centers in molecules. They possess one or more lone pairs of electrons or π-electrons and can donate an electron pair to form a covalent bond. Nucleophiles may be negatively charged species such as OH^-, CN^-, and Cl^-, or neutral molecules like NH_3 and H_2O that have lone pairs of electrons.

(Correct explanation with examples gives full marks)

13. Why do real gases behave ideally at high temperatures?

Negligible intermolecular forces: At high temperatures, gas molecules have high kinetic energy and move rapidly, overcoming the intermolecular attractions. As a result, these forces become insignificant, and the molecules behave as if they do not interact with each other.

Negligible particle volume: Although gas molecules have finite volume, their rapid motion at high temperatures makes the actual volume of the particles negligible compared to the total volume of the container. Hence, under high-temperature conditions, real gases closely follow the assumptions of the ideal gas model.

(correct explanation - give full marks)

14. What are primary and secondary standards in titrimetric analysis? Give example of each.

Primary standards are substances of very high purity, definite composition, and high stability. They are non-hygroscopic, do not react with air, and have relatively high molar mass, which minimizes weighing errors. Because of these properties, they can be weighed accurately to prepare a solution of known concentration directly. *Examples:* Sodium carbonate (Na_2CO_3), potassium dichromate ($K_2Cr_2O_7$).

Secondary standards are substances that are not sufficiently pure or stable to prepare standard solutions by direct weighing. Their solutions may be hygroscopic, volatile, or react with atmospheric gases (like CO_2), so their exact concentration must be determined by titration against a primary standard. *Examples:* Sodium hydroxide (NaOH), hydrochloric acid (HCl). Thus, primary standards are used to standardize secondary standards, ensuring accuracy in titrimetric analysis.

(Correct explanation with examples gives full marks)

15. Explain anti-Markovnikov addition with a suitable example.

Anti-Markovnikov (peroxide effect or Kharasch effect) is a type of addition reaction in which the substituent (such as HBr) adds to an unsymmetrical alkene in the opposite manner to Markovnikov's rule. Here, the hydrogen atom attaches to the carbon with fewer hydrogen atoms, while the halide (Br) attaches to the carbon with more hydrogen atoms. This occurs in the presence of peroxides via a free radical mechanism.

Example**:** Addition of HBr to propene in the **presence of peroxide:**

$CH_3–CH=CH_2 + HBr \rightarrow CH_3–CH_2–CH_2Br$ (1-bromopropane)

(Explanation with relevant examples give full marks)

16. What is the main difference between solids, liquids, and gases?

In **solids**, particles are closely packed with **strong intermolecular forces**, restricting them to vibrations only, giving a definite shape and volume. *Examples:* Sodium chloride (NaCl), copper (Cu), diamond.

In **liquids**, particles are close together with **moderate intermolecular forces**, allowing them to move past each other, so they have definite volume but no fixed shape. *Examples:* Water (H_2O), ethanol (C_2H_5OH), benzene (C_6H_6).

In **gases**, particles are far apart with **very weak or negligible intermolecular forces**, moving freely, resulting in neither definite shape nor definite volume. *Examples:* Oxygen (O_2), nitrogen (N_2), carbon dioxide (CO_2).

(Correct explanation with relevant examples gives full marks)

[3x4 = 12]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

17. Describe the principle of Redox titration with a suitable example.

Redox titration is an analytical technique used to determine the concentration of a substance based on an oxidation–reduction reaction involving the transfer of electrons. In this process, a solution of known concentration, called the titrant, is added to a solution of unknown concentration, called the analyte, until the reaction reaches completion.

The principle of redox titration is based on the transfer of electrons between the oxidizing agent and the reducing agent. Oxidation involves the loss of electrons, whereas reduction involves the gain of electrons. The equivalence point is reached when the number of electrons lost by one species is equal to the number of electrons gained by the other. The end point is usually detected by a colour change, either by using suitable indicators such as methylene blue or diphenylamine, or by the self-indicating property of certain titrants like potassium permanganate ($KMnO_4$).

Example: In the titration of ferrous sulphate ($FeSO_4$) with potassium permanganate ($KMnO_4$) in acidic medium, Fe^{2+} is oxidized to Fe^{3+} and MnO_4^- is reduced to Mn^{2+}. $KMnO_4$ acts as a self-indicator; the end point is indicated by the appearance of a permanent light pink colour.

$2\ KMnO_4 + 10\ FeSO_4 + 8\ H_2SO_4 \rightarrow 2MnSO_4 + 5Fe_2(SO4)_3 + K_2SO_4 + 8H_2O$

Mn^{+7} in $KMnO_4$ has been reduced to Mn^{+2} in $MnSO_4$. While Fe^{+2} in $FeSO_4$ has been oxidized to Fe^{+3} in $Fe_2(SO_4)_3$. Thus, $KMnO_4$ acts as oxidizing agent and $FeSO_4$ acts as reducing agent in acidic medium.

(Principle (5 marks): Define oxidation and reduction, identify oxidising and reducing agents, and explain the process in terms of electron transfer; Any one suitable example with a balanced chemical equation (2 marks)).

18. What are polymers? Illustrate addition polymerisation by taking polyethene and PVC as examples.

Polymers are high molecular weight macromolecules formed by the repeated linking of small units called **monomers** through covalent bonds. These repeating units create long chains (linear, branched, or cross-linked) with unique physical and chemical properties.

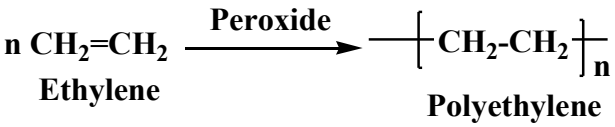
Addition Polymerisation

Addition polymerisation is a process in which unsaturated monomers (usually containing double bonds) combine together without the elimination of any small molecule. The reaction proceeds through chain mechanisms such as free radical, cationic, or anionic pathways.

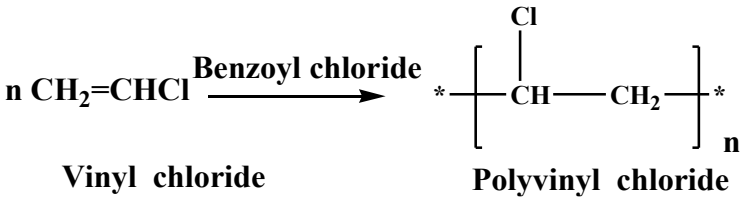
The process generally involves three steps:

Initiation – formation of reactive species (free radicals, cations, or anions)
Propagation – successive addition of monomer units to the growing chain
Termination – stopping of chain growth

For example, polyethylene can be made from chain (addition) polymerization of ethylene obtained by cracking of petroleum fraction,.**LDPE (Low Density Polyethene)** is produced at high pressure (1000 atm) and high temperature (180–250°C) using peroxide catalysts. **HDPE (High Density Polyethene)** is produced at low pressure (5–25 atm) and low temperature (25–50°C) using Ziegler–Natta catalysts.



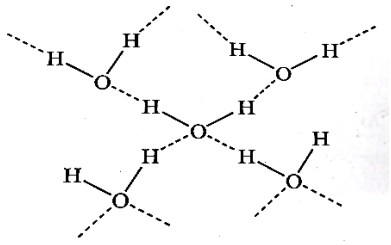
Polyvinyl Chloride is prepared by the addition polymerization of vinyl chloride in the presence of benzoyl chloride.



(Polymer: Definition – 1 mark; Explanation of addition polymerisation – 2 marks; Examples – 4 marks (2 + 2))

19. Describe the role of hydrogen bonding in determining the boiling and melting points of water. How do intermolecular hydrogen bonds contribute to the unique physical properties of water?

Water is a polar molecule in which the oxygen atom is more electronegative than hydrogen, resulting in partial negative charge on oxygen and partial positive charges on hydrogen atoms. This polarity enables the formation of **intermolecular hydrogen bonds** between water molecules. Each water molecule can form **four hydrogen bonds** two through its hydrogen atoms and two through the lone pairs of electrons on oxygen. This leads to the formation of a **three-dimensional tetrahedral network**.



Boiling Point of Water In water, molecules are held together by strong intermolecular hydrogen bonds. To convert liquid water into steam, these hydrogen bonds must be broken. Since hydrogen bonds are relatively strong, a large amount of energy (heat) is required to overcome them. Therefore, water has an unusually high boiling point (100°C) compared to similar molecules like H_2S, which do not form hydrogen bonds.

Melting Point of Water In ice, water molecules are arranged in a rigid three-dimensional hydrogen-bonded network. To melt ice, this structured arrangement must be partially broken. This requires significant energy to disrupt hydrogen bonds, resulting in a relatively high melting point (0°C).

Properties of water due to strong intermolecular attraction between water molecules (hydrogen bonding) give it exceptional thermal stability, surface properties, and solvent ability, making it vital for biological and environmental systems.

High specific heat capacity:
Water can absorb a large amount of heat without a significant rise in temperature because energy is used to break hydrogen bonds.

High heat of vaporization:
A large amount of energy is required to convert water from liquid to vapour due to the need to break intermolecular hydrogen bonds.

High surface tension:
Strong hydrogen bonding creates high cohesion between molecules, forming a tight surface layer.

Density anomaly (ice floats on water):
In ice, hydrogen bonds form an open 3D structure, making it less dense than liquid water.

Strong cohesion and adhesion:
Cohesion keeps water molecules together, while adhesion allows them to stick to other surfaces, aiding capillary action.
Excellent solvent properties:
Due to polarity and hydrogen bonding, water dissolves many ionic and polar substances effectively.

(Hydrogen Bonding – 1 mark; Structure of Water – 1 mark; Boiling and Melting Point Explanation – 3 marks; Any four Unique Physical Properties – 2 marks)

20. Define and discuss any four important concentration units commonly used in chemistry

Molarity (M)

*Number of moles of solute present in **1 litre of solution**.*

$$M = \frac{\text{moles of solute}}{\text{volume of solution (in litres)}}$$

Molality (m) (mol kg⁻¹)
Number of moles of solute present in **1 kg of solvent**.

$$m = \frac{\text{moles of solute}}{\text{mass of solvent (in kg)}}$$

Normality (N) (eq L⁻¹)
Number of **gram equivalents** of solute present in **1 litre of solution**.

$$N = \frac{\text{equivalents of solute}}{\text{volume of solution (in litres)}}$$

Mole Fraction (χ)
Ratio of moles of a component to the **total moles of all components** in the solution.

$$\chi_A = \frac{\text{moles of component A}}{\text{total moles of solution}}$$

Mass Percentage (%)
Mass of solute present in 100 parts by mass of solution.

$$\text{Mass \%} = \left(\frac{\text{mass of solute}}{\text{mass of solution}}\right) \times 100$$

Parts per Million (ppm)
Number of parts of solute present in one million parts of solution.

$$\text{ppm} = \left(\frac{\text{mass of solute}}{\text{mass of solution}}\right) \times 10^6$$

Parts per Billion (ppb)
Number of parts of solute present in one billion parts of solution.

$$\text{ppb} = \left(\frac{\text{mass of solute}}{\text{mass of solution}}\right) \times 10^9$$

(Definition and proper explanation of any four units will carry 7 marks (1.75* **x4**)*)

21. Illustrate the complete stepwise mechanism of E1 elimination, using curved-arrow notation. Discuss the role of carbocation stability and how reaction conditions influence the mechanism.

Elimination Reactions Unimolecular (E1)

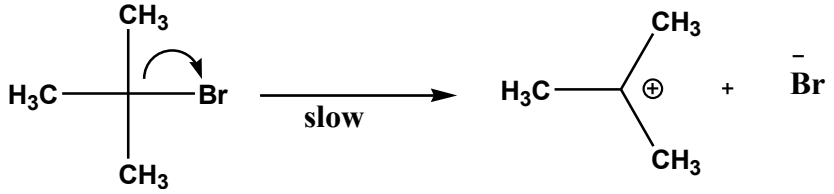
The **E1 reaction** (Elimination Unimolecular) is a two-step organic reaction mechanism where a leaving group departs to form a carbocation intermediate, followed by the loss of a proton to create a double bond. Usually follows **Zaitsev's Rule**, forming the more substituted (stable) alkene. It is commonly followed by tertiary alkyl halides due to the formation of a stable carbocation.

Mechanism

The mechanism occurs in two steps:

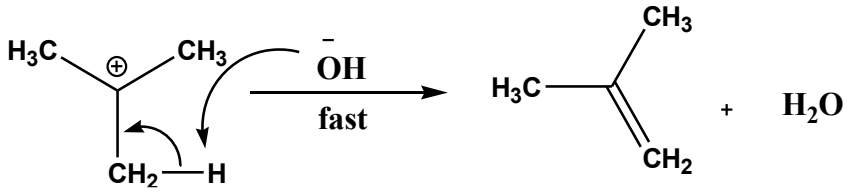
Step 1: Formation of Carbocation (Rate-Determining Step)

The carbon–halogen bond breaks heterolytically to form a carbocation and a halide ion.



Step 2: Deprotonation (Fast Step)

A base (such as OH^- or solvent) removes a proton from the β-carbon, leading to the formation of an alkene.



Since the rate depends only on the concentration of the alkyl halide, the reaction follows first-order kinetics:

$$\text{Rate} \propto [\text{alkyl halide}]$$

Carbocations are **electron-deficient, sp² hybridised species** with an empty p-orbital. They are stabilized by factors that donate electron density into this orbital. The stability order is:

Tertiary > Secondary >> Primary > Methyl.

due to the **+I effect** and **hyperconjugation** of alkyl groups. Therefore, tertiary substrates readily undergo E1 reactions, secondary substrates react moderately, while **primary substrates rarely form carbocations and instead undergo E2 reactions**.

Resonance-stabilized carbocations (allylic and benzylic) are highly stable and strongly favour the E1 pathway.

Reaction conditions also play an important role. **Polar protic solvents** (such as water and alcohols) stabilize both the carbocation and the leaving group through solvation, thereby lowering the energy required for ionization. A **good leaving group** (e.g., Cl^-, Br^-, I^-) facilitates carbocation formation, and **higher temperature** favours elimination

over substitution. Additionally, carbocations may undergo **rearrangements** (hydride or alkyl shifts) to form more stable intermediates before elimination.

(Mechanism with explanation – 4 marks; role of stability of carbocation – 1 mark, reaction conditions-2 marks)

22. Explain the concept of dipole-dipole interactions. How do these forces arise, and what factors influence their strength? Provide examples of substances where dipole-dipole interactions occur.

Dipole–dipole interactions are intermolecular forces of attraction that occur between polar molecules. These molecules possess a permanent dipole moment due to an unequal distribution of electrons between atoms with different electronegativities.

How these forces arise

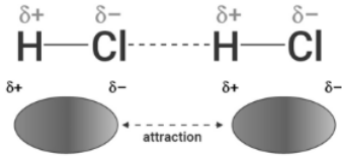
In a polar molecule, one end becomes partially positive (δ^+) and the other partially negative (δ^-). When such molecules come close:

The δ^+ end of one molecule is attracted to the δ^- end of another molecule

This electrostatic attraction forms a dipole–dipole interaction

For example:

In HCl, hydrogen carries a slight positive charge and chlorine carries a slight negative charge. Neighbouring HCl molecules align such that opposite charges attract.



Factors influencing their strength

Magnitude of dipole moment

Greater polarity → stronger dipole–dipole attraction

Molecules with larger electronegativity differences show stronger interactions

Distance between molecules

Shorter distance → stronger force (follows inverse relation with distance)

Molecular orientation

Proper alignment (positive end near negative end) maximizes attraction

Molecular size and shape

Larger molecules may have stronger overall intermolecular forces, but orientation effects still matter

Examples of substances with dipole–dipole interactions

Hydrogen chloride (HCl)

Sulfur dioxide (SO_2)

Chloroform ($CHCl_3$)

Acetone ($(CH_3)_2CO$)

(Definition / Concept – 2 marks, how these forces arise – 1 mark, factors influencing strength – 3 marks, examples of substances – 1 mark)

[7x4 = 28]